Boyer's Casimir Repulsion as Euler Product Truncation: Number-Theoretic Reinterpretation and Three Routes to Experimental Realization

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Boyer (1974) predicted that the Casimir force between a perfectly electrically conducting (PEC) plate and a perfectly magnetically conducting (PMC) plate is repulsive, in contrast to the attractive PEC–PEC case. This prediction remains unverified after fifty years, primarily because no PMC material exists at the relevant (optical–UV) frequencies. We offer a number-theoretic reinterpretation: the PEC–PMC (Dirichlet–Neumann) mode sum $1+3+5+7+\cdots = \zeta_{\neg 2}(-1) = +1/12$ is precisely the Riemann zeta function with the p=2 Euler factor removed, while the PEC–PEC sum $1+2+3+4+\cdots = \zeta(-1) = -1/12$ is the full zeta function. The sign flip is thus the arithmetic operation of *prime muting*: removing the contribution of the smallest prime from the vacuum mode sum. This reinterpretation clarifies why PMC is not required—any physical configuration selecting odd-integer modes suffices. We identify three such configurations that bypass the PMC obstacle: (A) nanoscale open-ended structures providing Neumann boundary conditions without magnetic materials; (B) $\lambda/4$ superconducting resonators in circuit QED, where Lamb shift differences between $\lambda/4$ and $\lambda/2$ resonators probe the vacuum energy sign; and (C) Si-MEMS devices with DN-type geometries, extending the platform of Tang *et al.* (2017). All three are realizable with existing technology. A positive result would constitute the first experimental verification of Boyer's fifty-year-old prediction and the first physical realization of Euler product truncation.

I. INTRODUCTION: A FIFTY-YEAR GAP

In 1974, Boyer showed that the Casimir energy between a perfect electric conductor (PEC, Dirichlet boundary condition for the electric field) and a perfect magnetic conductor (PMC, Neumann boundary condition) is *positive*—opposite in sign to the standard PEC–PEC result [1]. Kenneth *et al.* provided a rigorous proof [2], and Hertzberg *et al.* confirmed it in the piston geometry [3].

Despite this theoretical clarity, **no experiment has ever measured the PEC–PMC Casimir force**. The obstacle is material: PMC requires magnetic permeability $\mu \to \infty$ at the optical and ultraviolet frequencies that dominate the Casimir energy. No natural material satisfies this, and metamaterial approaches fail due to Kramers–Kronig constraints on broadband magnetic response [4].

Existing demonstrations of Casimir "repulsion" [5, 6] use different mechanisms: Munday *et al.* employed dielectric contrast in a fluid; Tang *et al.* used geometric interplay of T-shaped nanostructures. In neither case is the vacuum mode structure altered—the underlying mode sum remains the full $\sum_{n=1}^{\infty} n^{-s}$, and no modes are selectively removed. These experiments demonstrate force-direction control, not vacuum energy sign reversal.

II. NUMBER-THEORETIC REINTERPRETATION

The Riemann zeta function admits an Euler product over primes:

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s} = \prod_{p \text{ prime}} \frac{1}{1-p^{-s}}. \tag{1}$$

Removing the p=2 factor yields

$$\zeta_{\neg 2}(s) \equiv (1-2^{-s})\,\zeta(s) = \sum_{\substack{n=1 \\ n \text{ odd}}}^{\infty} n^{-s}. \tag{2}$$

At $s=-1$:

$$\zeta(-1) = 1+2+3+4+\cdots = -\tfrac{1}{12}, \tag{3}$$

$$\zeta_{\neg 2}(-1) = 1+3+5+7+\cdots = +\tfrac{1}{12}. \tag{4}$$

The sign flip $-1/12 \to +1/12$ is the *defining property* of prime-2 muting.

Boyer's result is precisely this:

- PEC–PEC (DD): modes $n = 1, 2, 3, \ldots \;\Rightarrow\; E \propto \zeta(-1) = -1/12 < 0$ (attractive).
- PEC–PMC (DN): modes $n = 1, 3, 5, \ldots \;\Rightarrow\; E \propto \zeta_{\neg 2}(-1) = +1/12 > 0$ (repulsive).

Boyer computed $\zeta_{\neg 2}$ without knowing it. His electromagnetic boundary-condition calculation and the number-theoretic Euler product truncation are the same mathematical operation, viewed from different angles.

This reinterpretation has a crucial practical consequence: *PMC is not needed.* Any physical configuration whose mode spectrum consists of odd integers realizes $\zeta_{\neg 2}$. The task is not to find a magnetic material, but to find a geometry that selects odd modes.

III. THE $\lambda/4$ REALIZATION

A $\lambda/4$ resonator—one end shorted (Dirichlet), the other open (Neumann)—has mode frequencies

$$f_n = (2n-1)\,\frac{c}{4L}, \quad n = 1, 2, 3, \ldots \tag{5}$$

These are the odd integers $1, 3, 5, 7, \ldots$ in units of $c/(4L)$. The vacuum energy is

$$E_{\lambda/4} = \frac{\pi\hbar c}{4L}\,\zeta_{\neg 2}(-1) = +\frac{\pi\hbar c}{48L} > 0. \tag{6}$$

Compare the $\lambda/2$ resonator (both ends shorted or both open):

$$E_{\lambda/2} = \frac{\pi\hbar c}{2L}\,\zeta(-1) = -\frac{\pi\hbar c}{24L} < 0. \tag{7}$$

Opposite signs. No PMC material. No metamaterial. No Kramers–Kronig obstruction. The open end of a transmission line provides a broadband, dispersion-free Neumann boundary condition for TEM modes.

The obstacle is scale: transmission-line $\lambda/4$ resonators are macroscopic ($\sim$cm), where the Casimir force $\sim 10^{-21}$ N is unmeasurable. We propose three routes around this obstacle.

IV. THREE ROUTES TO EXPERIMENTAL REALIZATION

A. Route A: Nanoscale DN Structures

At the nanoscale ($a \sim 100$ nm), Casimir forces are measurable ($\sim$pN). The challenge is to create DN boundary conditions at this scale.

A Neumann condition for the electromagnetic field corresponds physically to an open aperture, a slot, or a free edge. Candidate structures include:

- Open-ended metallic nanowaveguides (slot arrays in thin metal films).
- Nanoscale coaxial structures with open terminations, fabricable by template-based electrodeposition [7].
- Photonic crystal surfaces engineered to exhibit high-impedance (PMC-like) behavior over a broad bandwidth.

The Casimir force between a PEC plate and such a nano-DN surface can be measured with standard AFM techniques.

B. Route B: Circuit QED ($\lambda/4$ vs. $\lambda/2$)

In circuit QED, a transmon qubit coupled to a superconducting resonator experiences a Lamb shift

$$\delta f = \sum_n \frac{g_n^2}{f_q - f_n}, \tag{8}$$

where f_n are the resonator mode frequencies and g_n the coupling strengths.

A $\lambda/4$ resonator (modes $1, 3, 5, \ldots$) and a $\lambda/2$ resonator (modes $1, 2, 3, \ldots$) of the same fundamental frequency will produce *different* Lamb shifts, because the mode sums in (8) differ. The difference

$$\Delta(\delta f) = \delta f_{\lambda/2} - \delta f_{\lambda/4} = \sum_{n \text{ even}} \frac{g_n^2}{f_q - f_n} \tag{9}$$

isolates the even-mode vacuum fluctuation contribution.

Both $\lambda/4$ and $\lambda/2$ resonators are standard components in superconducting circuit laboratories. The measurement requires a dilution refrigerator ($\sim$10 mK) and a transmon qubit—existing infrastructure at dozens of laboratories worldwide.

C. Route C: MEMS with DN Geometry

Tang *et al.* [6] demonstrated on-chip Casimir force measurement between silicon nanostructures using integrated comb actuators and vibrating-beam force sensors. Their T-shaped geometry produced non-monotonic (but not genuinely DN) forces.

We propose adapting their platform with a DN-type geometry: one surface is a flat PEC (gold-coated silicon), the other is a periodic array of open-ended slots or channels whose mode structure approximates DN boundary conditions. The force is measured as a function of separation a using the same comb-actuator technology.

The key measurement is the scaling exponent α in $F \propto a^{-\alpha}$. Standard PEC–PEC gives $\alpha = 5$ (attractive, well verified). PEC–DN should give $\alpha = 5$ with *opposite sign* (repulsive, never measured).

V. PREDICTIONS

1. **Sign**: The Casimir force in a DN configuration is repulsive. This is Boyer's prediction, now reinterpreted as Euler product truncation $\zeta \to \zeta_{\neg 2}$.
2. **Magnitude**: In 3D, the ratio of DN to DD Casimir energy densities is $\zeta_{\neg 2}(-3)/\zeta(-3) = -7$. The DN repulsive force is 7 times the magnitude of the DD attractive force at the same separation.
3. **Scaling**: Standard QED predicts $F \propto a^{-5}$ for both DD and DN. Any deviation from $\alpha = 5$ in the DN case would indicate physics beyond the standard boundary-effect framework.

VI. DISCUSSION

A. Why This Hasn't Been Done

The PEC–PMC Casimir repulsion has remained unverified for fifty years because the problem was framed as a *materials* problem ("find a PMC"). The number-theoretic reinterpretation reframes it as a *mode selection* problem ("select odd modes"), which can be solved geometrically.

The $\lambda/4$ solution—simply opening one end of a resonator—is arguably the simplest realization of Boyer's prediction. It has not been proposed in this context because (i) $\lambda/4$ resonators are macroscopic where Casimir forces are negligible, and (ii) the connection between DN modes and Euler product truncation was not previously recognized.

B. Relation to Broader Program

The Euler product $\zeta(s) = \prod_p (1-p^{-s})^{-1}$ decomposes the vacuum into prime-indexed channels. Removing the p=2 channel ($\zeta \to \zeta_{\neg 2}$) is the simplest nontrivial modification. The sign flip $\zeta(-1) \to \zeta_{\neg 2}(-1)$ $(-1/12 \to +1/12)$ and $\zeta(-3) \to \zeta_{\neg 2}(-3)$ $(+1/120 \to -7/120)$ suggest that prime muting could, in principle, reverse the sign of the cosmological vacuum energy—a necessary condition for exotic spacetime geometries [8, 9]. Experimental verification of Boyer's sign flip in any of the three proposed routes would be the first step toward testing this broader hypothesis.

C. What This Paper Does Not Claim

We do not claim that prime muting is physically realizable beyond the DN boundary condition. The connection between DN modes and the p=2 Euler factor is a mathematical identity, not a physical mechanism. Whether the arithmetic structure of the Euler product has deeper physical significance is an open question that experiment must address.

VII. CONCLUSION

Boyer's 1974 prediction that PEC–PMC Casimir force is repulsive can be reinterpreted as the statement that removing the p=2 Euler factor from the Riemann zeta function reverses the sign of the regularized vacuum energy. This reinterpretation shifts the experimental challenge from materials (finding a PMC) to geometry (selecting odd modes). We have identified three experimentally viable routes—nanoscale DN structures, circuit QED $\lambda/4$ comparison, and MEMS with DN geometry—each realizable with existing technology. A positive result would simultaneously verify a fifty-year-old theoretical prediction and provide the first physical realization of Euler product truncation in quantum field theory.

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